

Pilot Setup

A 12-week automation pilot was conducted at the Rotterdam distribution center of a major grocery retailer, deploying 40 Atlas P-100 robots across two warehouse zones.

The pilot ran from March through May 2025.

Methodology

The study used an A/B design: Zone A operated with robot assistance while Zone B continued manual picking as a control. The primary metric was picks per hour per worker.

Error rates were measured as mis-picks per 1,000 order lines.

Results

Zone A throughput increased by 34% compared to the control zone. The mis-pick error rate fell from 1.8% to 0.4% over the pilot period.

Average order cycle time dropped from 42 minutes to 28 minutes.

Incidents and Reliability

Three minor stoppages occurred during the pilot, all resolved within 20 minutes. No safety incidents involving personnel were recorded.

Mean time between failures (MTBF) across the fleet was 410 hours.

Conclusions

The projected return on investment period is 14 months. The study recommends rolling out the system to three additional distribution sites in 2026.

Worker satisfaction surveys showed a 12-point improvement after robot deployment.